USING A BIO-INSPIRED MODEL TO FACILITATE THE ECOSYSTEM OF DATA SHARING IN SMART HEALTHCARE

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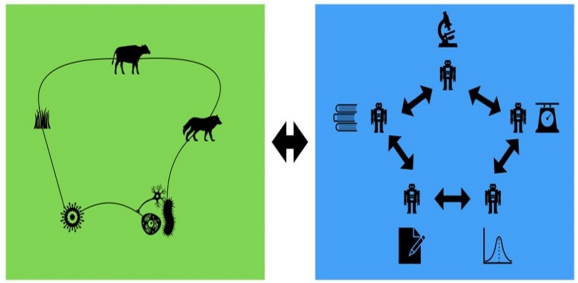
Abstract

Following the development of Information Technology (IT) techniques, data and the knowledge behind data have been increased exponentially today. The goals to better manage and share such massive big data become more and more critical in many prominent Artificial Intelligence (AI)-based smart industries. Smart healthcare is a typical example in these cases. Efficient management of data sharing apparently can lead to better diagnosis, illness prevention and epidemic monitoring. However, appropriate and robust management of sharing sensitive data among different stakeholders operating within the ecosystem still poses challenges of geographical boundaries and compliance to diverse data sharing and access rules across continents. To address the complexity of large-scale data sharing and provide an efficient solution, our study proposes a bio-inspired autonomic agent-based framework capable of leveraging the large-scale distributed data sharing infrastructure with multiple stakeholders whilst, at the same time, supporting the development of future data integration ecosystems. Our current study selects smart healthcare as a use case to discuss the deployment of our data sharing framework and use this example to demonstrate the potential advantage of this framework but the framework itself also has good extensibility to other interdisciplinary scenarios and domains.

Key words

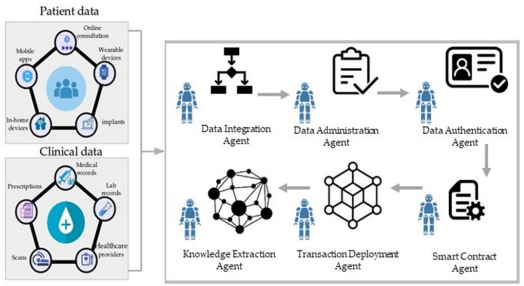
Bio-inspired Agents, Evolutionary Computation, Data Sharing

Bio-inspired Collective Agents Imitate The Food Chain Model In Nature



- Analogy between biological and business models
- ***Biological models***
  - Each organism represents a stakeholder
  - Interaction among the organisms adapt the changing environments
- ***Business models***
  - Each stakeholders will be represented by agent
  - Equilibrium in the network with the agents
- Adapting biological models in business augments productivity in business models

Self-organizing Data Sharing Framework And Dynamic Regulation In Smart Healthcare



- ***Data Integration***
  - Data sources
    - Patient data – wearable devices, smart phones, etc.
    - Clinical data – diagnosis and lab reports, etc.
- ***Data administration***
  - Updating kowldege with patient's latest vists to hospitals
- ***Smart Contracts for authentication and authorization***
  - Security check-ups with smart contracts in blockchain
- ***Transaction broadcast in blockchain network***
  - Decentralizing the transaction to ensure security
- ***Agents to automate these processes***
  - Evolutionary models to adapt the agents

Conclusion

This article described how to use computational bio-inspired agents to access the different distributed datasets and keep the data and knowledge sharing efficient, robust and compatible with all stakeholders in a smart healthcare system. This proposed framework provides a potential interdisciplinary solution for data sharing on many domains where it may need to implement intensive collaboration based on common data. Like the analogy of the natural food chain model, our proposed framework also inherits similar advantages to its biological counterpart. It is capable of dynamically adapting to the customized requests of various users and reaching a consensus through massive interaction. In the next step of this study, we will continually test the framework in a series of subsequent scenarios and initialize the interdisciplinary collaboration with the experts from different domains. Within the future collaboration, we will develop the corresponding benchmark tests based on the expert's suggestion to examine the functionality of the framework.

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